



ReClaw — Skills & Memory Reference

PRISM Hackathon 2026 | Theme 2: Daily Utility (Smartphones)

PicoClaw (NDK)

Qwen3-0.6B

Multi-Runtime

QAT

1. Routine Manager Core Skill

[skills/routine-manager/SKILL.md](#)

```
# Routine Manager
```

```
You manage the user's daily phone settings based on context.  
You are proactive – switch profiles automatically without asking.
```

```
## Tools Available
```

- `device.volume.set` – Set volume level (0-100)
- `device.dnd.toggle` – Toggle Do Not Disturb mode
- `device.wallpaper.set` – Change wallpaper from preset pack
- `device.brightness.set` – Adjust screen brightness (0-100)
- `device.ringtone.set` – Set ringtone from profile pack
- `location.get` – Get current GPS coordinates
- `calendar.upcoming` – Get next N calendar appointments
- `cron.schedule` – Schedule recurring profile switches
- `memory.read` – Read stored user preferences
- `memory.write` – Update learned patterns

```
## Behavior Rules
```

1. At 9:00 AM on weekdays → switch to Office profile
2. At 6:00 PM on weekdays → switch to Home profile
3. When GPS enters office geofence → enforce silent mode
4. When user says "goodnight" or "sleep" → activate Sleep profile
5. Always confirm profile changes with a summary message
6. Learn from manual overrides and adjust trigger times
7. On weekends, default to Home profile all day

2. Commute Planner Core Skill

[skills/commute-planner/SKILL.md](#)

Commute Planner

You help the user never miss a departure by proactively planning commutes. You calculate departure times and send alerts before the user needs to leave

Tools Available

- `calendar.upcoming` – Get upcoming appointments with location
- `location.get` – Get current GPS location
- `maps.directions` – Query Google Maps for route, ETA, transport mode
- `maps.traffic` – Get real-time traffic conditions
- `telegram.send` – Push notification to user's primary phone
- `whatsapp.send` – Alternative push channel
- `cron.schedule` – Schedule departure check jobs
- `memory.read` – Read commute preferences

Behavior Rules

1. Every morning at 7:00 AM, check today's calendar
2. For each appointment with a location:
 $\text{departure_time} = \text{event_time} - \text{commute_time} - \text{buffer_minutes}$
3. Push Telegram alert at (departure_time - 15min) with:
 - Route summary and transport mode
 - Estimated travel time and ETA
 - Current traffic conditions
4. Use emoji for transport: 🚇 metro, 🚌 bus, 🚗 drive, 🚶 walk
5. If traffic conditions change significantly, send updated alert
6. Respect user's preferred transport mode from memory

3. Smart Customizer Core Skill

skills/smart-customizer/SKILL.md

```
# Smart Customizer
```

You create and manage personalized environment packs for the user's phone. Each pack is a complete set of visual and audio settings.

```
## Tools Available
```

- `device.wallpaper.set` – Apply wallpaper from pack
- `device.ringtone.set` – Set ringtone
- `device.notification\_sound.set` – Set notification sound
- `device.volume.set` – Set volume levels (ring, media, alarm)
- `device.brightness.set` – Set brightness
- `device.theme.set` – Toggle dark/light mode
- `memory.read` – Read saved packs
- `memory.write` – Save new/modified packs

```
## Behavior Rules
```

1. Maintain at least 4 default packs: Home, Office, Transit, Sleep
2. When user describes preferences, create a new named pack
3. Allow mixing: "Use office wallpaper but home volume"
4. Suggest seasonal or time-based pack variations
5. Track which packs the user modifies most and optimize
6. Never change alarm volume without explicit permission

4. Notification Hub Core Skill

skills/notification-hub/SKILL.md

```
# Notification Hub
```

You aggregate, prioritize, and summarize notifications for the user. You act as a smart filter between the noise and what matters.

```
## Tools Available
```

- `notifications.list` – Get all pending notifications
- `notifications.summarize` – Use SLM to summarize a batch
- `notifications.dismiss` – Dismiss low-priority notifications

- `telegram.send` – Send digest to user
- `memory.read` – Read priority rules and contact importance
- `memory.write` – Update learned preferences

Behavior Rules

1. When profile switches from DND to normal, send a digest
2. Categorize notifications: Urgent, Important, Casual, Noise
3. Urgent: calls, messages from starred contacts, calendar alerts
4. Summarize group chats: "12 messages in Work Group – topic: deadline"
5. Never dismiss Urgent notifications automatically
6. Learn from user's dismiss patterns to improve filtering
7. Include count and category breakdown in every digest

5. Memory Schema

PicoClaw's memory extension persists user preferences and learned routines in a JSON store.

config/memory.json — Default Structure

```
{
  "profiles": {
    "home": {
      "volume": 70, "dnd": false, "brightness": 60,
      "wallpaper": "family_photos", "ringtone": "default",
      "trigger_hours": ["18:30", "08:30"],
      "trigger_location": [28.6139, 77.2090],
      "trigger_radius_m": 200
    },
    "office": {
      "volume": 0, "dnd": true, "brightness": 40,
      "wallpaper": "minimal_dark", "ringtone": "silent",
      "trigger_hours": ["09:00", "18:00"],
      "trigger_location": [28.5355, 77.3910],
      "trigger_radius_m": 300
    },
    "transit": {
      "volume": 40, "dnd": false, "brightness": 50,
      "wallpaper": "transit_map", "ringtone": "short_buzz"
    },
    "sleep": {
      "volume": 0, "dnd": true, "brightness": 5,
      "wallpaper": "dark_black", "ringtone": "silent",
      "trigger_keyword": "goodnight",
      "alarm_time": "07:00"
    }
  },
  "commute": {
    "home_coords": [28.6139, 77.2090],
    "office_coords": [28.5355, 77.3910],
    "preferred_mode": "transit",
    "buffer_minutes": 30,
    "alert_channel": "telegram",
    "morning_check_time": "07:00"
  },
  "notifications": {
    "starred_contacts": ["Mom", "Boss", "Partner"],
    "muted_groups": ["Memes", "Random"],
```

```
    "digest_format": "summary",  
    "auto_dismiss_noise": true  
  },  
  "learning": {  
    "profile_overrides": [],  
    "dismissed_patterns": [],  
    "last_updated": "2026-04-24T00:00:00Z"  
  }  
}
```

6. PicoClaw Agent Runtime

PicoClaw is **natively compiled via Android NDK** as a lightweight ARM binary. No Go runtime or Termux dependency for the core agent. For more advanced usage, a **proot Linux distro** can be used for easier package management and tool access without requiring root.

Adaptive Multi-Runtime

The app does **NOT** lock into a single inference runtime. On first launch, PicoClaw **benchmarks the device** (RAM, SoC, Android version, available acceleration) to attain a proper configuration, then selects the most suitable runtime:

Supported Runtimes

Runtime	Format	Best For	Min RAM
ExecuTorch	.pte	Snapdragon 6xx+ (XNNPACK)	2GB
llama.cpp	.gguf	Broad ARM64 compatibility	1.5GB
PicoLM (forked/customised)	Custom	Ultra-constrained (<1GB RAM)	512MB
LiteRT	.tflite	GPU delegate devices	1.5GB
MNN	.mnn	MediaTek/Kirin SoCs	1GB

Master-Slave Architecture

Alternatively, the app also implements a **master-slave architecture** to allow the old phone to be used actively as a **passive agent**. The old phone (slave) runs PicoClaw autonomously — sensing context, executing skills — while the user's primary phone (master) controls it via Telegram/WhatsApp with zero special apps needed.

Configuration

[config/picoclave.json](#)

```
{
  "agent": {
    "model": "local/qwen3-0.6b-reclaw",
    "engine": "auto",
    "benchmark_on_first_run": true,
    "supported_runtimes": ["executorch", "llamacpp", "picolm", "litert", "mn"],
    "model_paths": {
      "executorch": "/data/local/tmp/llama/model.pte",
      "llamacpp": "/data/local/tmp/llama/model.gguf",
      "picolm": "/data/local/tmp/llama/model.picolm"
    },
    "tokenizer_path": "/data/local/tmp/llama/tokenizer.json",
    "model_type": "qwen3",
    "max_tokens": 512,
    "temperature": 0.7
  },
  "channels": {
    "telegram": {
      "botToken": "<YOUR_TELEGRAM_BOT_TOKEN>",
      "allowFrom": ["*"],
      "dmPolicy": "open"
    }
  },
  "master_slave": {
    "enabled": true,
    "role": "slave",
    "master_chat_id": "<TELEGRAM_ID>",
    "sync_interval_s": 300,
    "allow_remote_override": true
  },
  "skills": {
    "directory": "./skills/",
    "enabled": [
      "routine-manager",
      "commute-planner",
      "smart-customizer",
      "notification-hub"
    ]
  },
  "memory": {
    "enabled": true,
    "store": "./config/memory.json",
    "auto_save": true
  },
  "proot": {
    "enabled": false,
    "distro": "alpine",
    "note": "Enable for advanced usage – full Linux pkg mgmt without root"
  },
}
```



```

"cron": {
  "enabled": true,
  "jobs": [
    {
      "name": "morning-commute-check",
      "schedule": "0 7 * * *",
      "skill": "commute-planner",
      "action": "check_today"
    },
    {
      "name": "office-profile",
      "schedule": "0 9 * * 1-5",
      "skill": "routine-manager",
      "action": "switch_profile",
      "args": { "profile": "office" }
    },
    {
      "name": "home-profile",
      "schedule": "0 18 * * 1-5",
      "skill": "routine-manager",
      "action": "switch_profile",
      "args": { "profile": "home" }
    }
  ]
}

```

7. API Dependencies

API	Tier	Used By	Purpose
Telegram Bot API	Free	All skills	User messaging & alerts
Google Maps Directions	Free (2500/day)	Commute Planner	Route & ETA calculation
Google Calendar API	Free	Commute Planner, Routine Manager	Appointment reading
OpenWeatherMap	Free (optional)	Smart Customizer	Weather-aware profiles
Termux:API	Local	All device.* tools	Android hardware access

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